STM v2 at UC San Diego: procurement budget (lean scenario)

Prepared for Prof. Oscar Vasquez · prices checked 21 October 2026 · all figures USD · companion spreadsheet STM_v2_budget.xlsx (totals are live formulas)

Summary

This is the purchase list for a second scanning tunnelling microscope, built at UCSD. The target is atomic resolution on HOPG in air; the minimum bar is stable tunnelling. It assumes Pt/Ir tip wire and HOPG come from UCSD labs, the whole structure is 3D printed with no machining, the electronics repeat v1's JLCPCB order (two controller boards and two preamps, fully placed), and money is spent only on what a lab doesn't readily have: the piezo and its driver, batteries and other specialty parts.

The total is $1,867.09, which is $132.91 under the $2,000 ceiling. That includes shipping, tariffs (35% on the JLCPCB order is the biggest), California tax and a 10% contingency. Coarse approach stays a stepper motor on a fine screw, as Jacob decided on 21 October; the stick-slip slider priced earlier is dropped. The earlier full-build scenario (metal, machined, tips and HOPG bought) came to $3,006.93; it is not included here.

Of the $1,112.07 in parts, $405.12 is current prices read from vendor or aggregator pages on 21 October, $517.58 is v1's invoice prices (almost all the JLCPCB repeat order), and $189.37 is estimates, mainly the high-voltage driver, which has not been designed. Every purchased line links to where it would be bought; four lines explain why no exact product page exists.

What the budget buys

- **Piezo.** A PiezoDrive TB3006-N tube ($131), as Prof. Vasquez suggested (a suggestion, not a confirmed diagnosis of v1), with a stepper-driven coarse approach: an Adafruit NEMA-17 stepper ($14), a TMC2209 microstepping driver ($8.95) and three 1/4-80 fine screws ($8.78 each, v1 price). The tube needs high-voltage drive, so a provisional DIY driver is priced: six OPA454 amplifiers, a transformer, rectifier and regulator parts, and a small PCB. Commercial PDu150 modules would add about $734 instead.
- **Electronics.** A repeat of the 2026-07-15 JLCPCB order: two fully assembled controller boards, two fully assembled preamps and the spare bare boards, at v1's invoice prices. JLCPCB only quotes after gerbers and a BOM are uploaded, so it could not be re-quoted; part prices have moved since. Plus a Teensy 4.1 and the separately bought LT1469.
- **Preamp changes from Baird.** A 1 GΩ feedback resistor, PTFE standoffs for the input node, lithium 9 V batteries and holders, and ±0.1 °C temperature sensors.
- **Printed structure and approach.** One 1 kg spool of filament (Prusament PETG-CF priced; nylon-CF is the alternative, choice open). The v1 rubber-band and spring plate is replaced by a rigid printed plate with a magnetic steel sample disc. The stepper turns one 1/4-80 screw (replacing v1's geared 28BYJ-48 and ULN2003); two hand-set screws and steel-ball pivots complete the tripod, on Sorbothane isolation.

What is not in the total

Everything on the In-kind and campus table: tip wire and HOPG (reference value about $354 if bought), granite or breadboard, silver epoxy, bench supply, scope, wire, connectors, fasteners, consumables, tools and enclosures. These are assumed, not confirmed. Printer

time and student labour are also excluded. Dr. Zahl advised against plastics in the mechanical loop; the printed structure goes against that, and it is flagged as an open risk.

Limits of this research

v1 actually cost $1,375.93 from receipts and the v1 purchase log. DigiKey, Mouser and McMaster-Carr block this environment, so their prices come from aggregator listings and are labelled that way. Thorlabs and Amazon product pages did not load. The UCSD mailbox and the Amazon account used for v1 were not reachable.

Budget summary

Line	Amount	Notes
1 Piezo scanner	$131.00	
2 Coarse approach (stepper on a fine screw)	$49.29	
3 Circuit boards from JLCPCB (assembled and bare)	$497.06	
4 Parts bought separately and hand-soldered onto the boards	$132.51	
5 Power: transformer and batteries	$117.73	
6 Modules wired in (microcontroller and sensors)	$54.50	
7 Printed structure, sample plate and isolation	$129.98	
Purchased components	$1,112.07	Equals the Detailed BOM subtotal
of which verified current prices	$405.12	
of which v1 invoice prices (mostly the JLCPCB repeat order)	$517.58	
of which estimates	$189.37	
Vendor shipping	$214.26	
Tariffs and import duties	$219.84	JLCPCB 35% is the largest piece
Carrier import processing fee	$65.00	
Sales and use tax (7.75%)	$86.19	
Subtotal before contingency	$1,697.36	
Contingency (10%)	$169.74	
TOTAL REQUEST (lean scenario)	**$1,867.09**	
Ceiling	$2,000.00	
Headroom (negative = over)	$132.91	
Check: total vs Vendor-order totals x 1.10	$0.00	Should be 0.00
Options and their effect		
Stick-slip approach instead of the stepper (not chosen), merchandise	$115.00	
Piezo option P2: TB3006-N + 2x PiezoDrive PDu150 + 2x PS1 supply	$733.80	
Piezo option P3: TB3006-N + PiezoDrive TD250-INV (vendor's recommended tube driver)	$3,013.80	
Piezo option P4: TB2006-N (20 mm, stiffer) + DIY driver	-$20.00	
Piezo option P5: TB4006-N (40 mm, more range) + DIY driver	$25.00	
Piezo option P6: Baseline + PiezoDrive factory wiring service	$100.00	
If campus can't lend silver epoxy (MG 8331D-14G)	$133.13	Reference value, In-kind tab
If campus can't provide a bench supply (v1 paid)	$42.74	
Earlier scenario, for comparison		
Full build (metal, machined, bought tips and HOPG, hand-assembled boards)	$3,006.93	From STM_v2_budget_full_build.xlsx (2026-10-21, first revision)
Full build after its four cuts	$2,626.74	

Line	Amount	Notes
Not in the total: everything on the In-kind and campus tab (assumed free, unverified), student labour, printer time.		

Unresolved items

ID	What is unknown	Current treatment	Effect on the total	What would resolve it
U1	**HV driver design**. Not drawn; only the op-amps are priced exactly	HV-01..04: $158.20 incl. transformer placeholder	Could move +/-$50; commercial PDu150 modules instead add about $734	Draw the schematic, then itemize with links
U2	**JLCPCB price today**. Can't re-quote without uploading gerbers and BOM	v1 unit prices + today's 35% tariff	Part prices have moved; likely +/-10% on $497	Upload the v1 files to JLCPCB for a quote
U3	**Campus items really free**. Printers, supply, scope, epoxy, wire, stock not confirmed	$0	Bench supply ~$43+, epoxy $133 if a lab can't lend	Confirm with Prof. Vasquez's lab
U4	**Printed structure vs Zahl's advice**. Zahl: no plastics in the mechanical loop	Printed (Jacob's choice)	Metal parts would bring back machining or stock cost	Decide with the advisers; ceramic/metal inserts at the tip-sample loop
U5	**Filament choice**. PETG-CF vs nylon-CF	PETG-CF 1 kg $54.99	Nylon-CF is similar or a bit more	Pick based on the campus printer
U6	**Screw bushings**. v1 ran the 1/4-80 screws in printed threads, which wear	None priced	Brass 1/4-80 bushings would add a few dollars each	Price Thorlabs or Newport bushings when their pages load
U7	**PiezoDrive shipping and duty**. No rate shown; DAP	$60 + 10% + $65 carrier fee	+/- $50	Ask PiezoDrive for a delivered quote
U8	**Sorbothane price**. Variant price not displayed	$35 estimate	+/- $10	Check the cart

Piezo purchasing options

Option	Description	Tube price	Driver / dependencies	Driver cost	Extra services	Option total	Change vs P1	Est. XY (um)	Est. Z (um)	Notes
P1	Baseline: TB3006-N + DIY 5-channel OPA454 driver at +/-45 V	$131.00	DIY OPA454 linear driver (HVD-01..07, provisional)	$158.20	$0.00	$289.20	$0.00	~5.0	~1.1	Lowest cost. Range scales with voltage: about 29*(45/260)=5.0 um XY and 6.3*(45/260)=1.1 um Z if linear (estimate from PiezoDrive's spec at +/-260 V). Driver is an undesigned new circuit.
P2	TB3006-N + 2x PiezoDrive PDu150 + 2x PS1 supply	$131.00	2x PDu150 ($413) + 2x PS1 ($33)	$892.00	$0.00	$1,023.00	$733.80	~13-16 (est.)	~3 (est.)	Commercial, low noise (26 uV). Output is -30 to +150 V (unipolar), 3 channels per module, so 5 tube channels need two modules. PiezoDrive does not state how to wire it to a quartered tube. One PS1 per module assumed.
P3	TB3006-N + PiezoDrive TD250-INV (vendor's recommended tube driver)	$131.00	TD250-INV	$3,172.00	$0.00	$3,303.00	$3,013.80	29 (spec)	6.3 (spec, half if Z shares)	PiezoDrive's own recommendation for tube scanners. Alone it exceeds the $2,000 ceiling.
P4	TB2006-N (20 mm, stiffer) + DIY driver	$111.00	DIY OPA454 driver	$158.20	$0.00	$269.20	-$20.00	~2.3	~0.7	Shorter tube: less range (13 um XY spec), higher resonance. Saves $20.
P5	TB4006-N (40 mm, more range) + DIY driver	$156.00	DIY OPA454 driver	$158.20	$0.00	$314.20	$25.00	~8.8	~1.3	More XY range (51 um spec), lower resonance.
P6	Baseline + PiezoDrive factory wiring service	$131.00	DIY OPA454 driver	$158.20	$100.00	$389.20	$100.00	~5.0	~1.1	Horizontal Tube Wiring service $100 (vertical wiring price not loaded). Removes hand-soldering of electrode leads.

Coarse approach and sample plate scenarios

Chosen: stepper on a 1/4-80 screw = PZ-02 + PZ-03 + PZ-04	$49.29
Motor: NEMA-17 (Adafruit 324) with a TMC2209 driver at up to 1/256 microstepping, about 6 nm of screw travel per microstep before any lever	
Two hand-set 1/4-80 screws and the motor screw form the tripod, with steel-ball pivots (ME-04)	
Not chosen, for the record: piezo stick-slip slider	

Item	Part	Link	Qty	Unit	Extended	Status	Notes
Piezo stack	PiezoDrive SA050510	https://www.piezodrive.com/product/150v-5x5x10mm-piezo-stack-actuator-sa050510/	1	$68.00	$68.00	Verified - vendor page	Also needed a sawtooth HV channel
Sapphire contacts, 2 mm	Edmund Optics 43-642	https://www.edmundoptics.com/p/20mm-diameter-sapphire-ball-lens/5174/	3	$28.25	$84.75	Verified - vendor page	
Ground guide rod	OnlineMetals W1 drill rod 1/8 in	https://www.onlinemetals.com/en/buy/tool-steel-drill-rod-w1	1	$4.82	$4.82	Verified - vendor page	
Stick-slip parts total (merchandise, plus one OPA454 channel)					$164.29		
Difference vs the chosen stepper drive (positive = stick-slip costs more)					$115.00		
Sample plate: printed rigid plate + magnetic 12 mm steel disc (ME-05) on lab magnets; HOPG mounted with lab silver epoxy/paint (Baird) or clipped (Zahl's sketch). No springs or rubber bands.							

Detailed bill of materials (purchases only)

ID	Item	Part	Vendor / link	Qty	Unit	Extended	Evidence	Notes
1 Piezo scanner								
PZ-01	Piezo tube scanner, 30 mm x 6 mm, quartered electrodes	PiezoDrive TB3006-N	PiezoDrive https://www.piezodrive.com/product/tb3006-n-29um-range-piezo-tube-scanner/	1	$131.00	$131.00	Verified - vendor page	Direct product page. Prof. Vasquez's suggestion (better piezo, possibly a tube); not a confirmed diagnosis and not a guarantee of atomic resolution.
2 Coarse approach (stepper on a fine screw)								
PZ-02	Coarse-approach stepper motor, NEMA-17, 200 steps/rev	Adafruit 324 (12 V, 350 mA)	Adafruit https://www.adafruit.com/product/324	1	$14.00	$14.00	Verified - vendor page	Direct product page, in stock. Jacob's choice (2026-10-21): stay with a stepper-driven screw instead of stick-slip. Replaces v1's geared 28BYJ-48. Runs from a 12 V bench supply (campus).
PZ-03	Stepper driver, microstepping (up to 1/256)	Adafruit 6121 (TMC2209 breakout)	Adafruit https://adafruit.com/product/6121	1	$8.95	$8.95	Verified - vendor page	Direct product page, in stock. Replaces v1's ULN2003; step/dir from the Teensy. Fine microstepping gives about 6 nm of screw travel per microstep before any lever.
PZ-04	Fine-pitch adjuster screws, 1/4-80 (one motor-driven, two hand-set)	Thorlabs FAS200	Thorlabs https://www.thorlabs.com/thorproduct.cfm?partnumber=FAS200	3	$8.78	$26.34	Historical - v1 receipt	Same screws as v1 (three bought 2026-06-24 at $8.78). Thorlabs pages do not load from here, so this is v1's price. 1/4-80 is 317.5 um per turn. v1 ran them in printed threads; matching brass bushings are not priced (U6).
3 Circuit boards from JLCPCB (assembled and bare)								
EL-01	Controller board, fully assembled (v1 design)	JLCPCB SMT assembly, v1 controller design	JLCPCB https://jlcpcb.com/	2	$209.64	$419.28	Historical - v1 receipt	v1 invoice 2026-07-15 unit price. Re-quote not possible from here: JLCPCB prices only after gerber + BOM upload. Parts prices have moved since (e.g. LT1469 now $23.77 and out of stock at LCSC).
EL-02	Preamp board, fully assembled (rev 2 gerbers)	JLCPCB SMT assembly, v1 preamp rev 2	JLCPCB https://jlcpcb.com/	2	$30.17	$60.34	Historical - v1 receipt	v1 unit price for the rev 1 preamp. Rev 2 adds 4.7 uF 50 V 1812 caps, a few cents each.
EL-03	Bare controller PCBs (as in the v1 order)	JLCPCB 2-layer	JLCPCB https://jlcpcb.com/	3	$1.68	$5.04	Historical - v1 receipt	v1 invoice line.
EL-04	Bare preamp PCBs (as in the v1 order)	JLCPCB 2-layer	JLCPCB https://jlcpcb.com/	3	$0.80	$2.40	Historical - v1 receipt	v1 invoice line.
HV-04	HV driver bare PCB (rides the JLCPCB order)	JLCPCB 2-layer, 5 pcs	JLCPCB https://jlcpcb.com/	5	$2.00	$10.00	Estimated	Link is JLCPCB's ordering site; a PCB has no product page until gerbers are uploaded. $2 for five is from a Hack Club guide (undated).
4 Parts bought separately and hand-soldered onto the boards								
EL-05	U21 dual op-amp for the controller board (JLCPCB does not supply it)	Analog Devices LT1469IN8#PBF	DigiKey https://www.digikey.com/en/products/detail/analog-devices-inc/LT1469IN8-PBF/962892	2	$18.26	$36.52	Estimated	v1 bought 2 at DigiKey ($16.35). DigiKey page blocked here; price is Mouser's via an aggregator. URL from search results.
PA-01	1 GOhm feedback resistor, soldered across the op-amp pins	Ohmite SM104031007FE	DigiKey https://www.digikey.com/en/products/detail/ohmite/SM104031007FE/824190	2	$5.73	$11.46	Verified - aggregator listing	1 + 1 spare (it is the part Baird's change depends on). URL from search results; DigiKey blocks this environment, price via aggregator (1,673 in stock).
PA-02	PTFE standoff terminal for the air-wired input node	Keystone 11301	DigiKey https://octopart.com/part/keystone/11301	2	$2.09	$4.18	Historical - v1 receipt	NO DIRECT DIGIKEY URL FOUND (DigiKey 36-11301-ND); link is the Octopart listing (Mouser $2.03). v1 paid $2.09 at DigiKey.

ID	Item	Part	Vendor / link	Qty	Unit	Extended	Evidence	Notes
HV-01	High-voltage op-amp for the piezo driver board, one per channel (5 tube + 1 spare)	TI OPA454AIDDAR	LCSC https://www.lcsc.com/product-detail/C50636.html	6	$6.72	$40.35	Verified - vendor page	Direct product page, 241 in stock. The driver circuit is not designed yet.
HV-03	Piezo driver board parts: rectifier, filter caps, HV regulators, heatsinks, HV connectors	not designed	LCSC *no direct link (see notes)*	1	$40.00	$40.00	Estimated	NO PRODUCT LINK: parts not chosen because the circuit is not drawn. Reference prices seen: GBU4J $1.25, Nichicon UVZ2A102 $2.28, TL783 $3.80.
5 Power: transformer and batteries								
HV-02	Transformer for the +/-45-50 V rails	toroid, secondary not chosen	Jameco https://www.jameco.com/z/182M18-Hammond-Manufacturing-Toroidal-Transformer-120VA-50Hz-Dual-Secondary-36VCT-3-33A-Series-18V-6-66A-Parallel_2575671.html	1	$67.85	$67.85	Estimated	NO EXACT PRODUCT PAGE: the secondary voltage is not chosen. The link is the nearest verified part (Hammond 182M18, 2 x 18 V, $67.85), which is NOT the right voltage; used as a price placeholder.
PA-03	9 V lithium batteries for the analog supply (two give +/-9 V)	Energizer Ultimate Lithium L522	BatteryMart https://www.batterymart.com/p-energizer-la522-9volt-lithium-battery.html	2	$21.95	$43.90	Verified - vendor page	Direct product page, in stock. Alkaline 4-pack at Battery Junction is $18.95 but was out of stock.
PA-04	9 V battery holder	Keystone 1294	DigiKey https://octopart.com/part/keystone/1294	2	$2.99	$5.98	Verified - aggregator listing	NO DIRECT DIGIKEY URL FOUND; link is the Octopart listing (Newark $2.99, in stock). Bought in the DigiKey order.
6 Modules wired in (microcontroller and sensors)								
EL-06	Microcontroller	Teensy 4.1 (SparkFun DEV-16771)	SparkFun https://www.sparkfun.com/teensy-4-1.html	1	$31.50	$31.50	Verified - vendor page	Direct product page. Headers are campus stock. v1 paid $43.22 on Amazon.
PA-05	Temperature sensor, +/-0.1 C (head and preamp)	Adafruit 4821 (TI TMP117)	Adafruit https://adafruit.com/product/4821	2	$11.50	$23.00	Verified - vendor page	Direct product page. Zahl: monitor temperature; Baird: steady amplifier temperature.
7 Printed structure, sample plate and isolation								
ME-01	3D-printing filament for head, sample plate and approach body	Prusament PETG Carbon Fiber, 1.75 mm, 1 kg	Prusa Research https://www.prusa3d.com/product/prusament-petg-carbon-fiber-black-1kg/	1	$54.99	$54.99	Verified - vendor page	Direct product page, in stock. One spool covers the parts and reprints. Material is still open: PETG-CF is stiff and easy to print; nylon-CF is stiffer but needs a dried, hardened-nozzle printer. Zahl advised against plastics in the mechanical loop; printing is Jacob's choice for this budget.
ME-04	Hardened steel balls for kinematic pivots	uxcell 52100 G25, 0.18 in, 100 pcs	Walmart https://www.walmart.com/ip/533082272	1	$9.99	$9.99	Verified - vendor page	Direct product page.
ME-05	Magnetic stainless sample discs, 12 mm (HOPG carrier)	Ted Pella 16208, pack of 50	Ted Pella https://www.tedpella.com/AFM_html/AFM.aspx	1	$30.00	$30.00	Verified - vendor page	Catalogue page lists 16208 at $30.00 per 50. Replaces v1's rubber-band/spring plate.
ME-06	Sorbothane isolation pads, 0.5 in, 50 duro, 6x6 in	Sorbothane 0266050-50-10 (2 per pack)	Isolate It https://www.isolateit.com/products/sorbothane-acoustic-and-vibration-isolation-square-6-x-6-pads	1	$35.00	$35.00	Estimated	Direct product page; this variant's price was not displayed (only the 0.08 in pad, $24.95).

Vendor and order totals

Vendor	Merchandise	Shipping	Shipping basis	Tariff rate	Tariff / duty	Sales or use tax	Order total
PiezoDrive	$131.00	$60.00	Estimated	0.1	$13.10	$10.15	$214.25
Adafruit	$45.95	$6.00	Estimated	0	$0.00	$3.56	$55.51
Thorlabs	$26.34	$15.26	Historical - v1 receipt	0	$0.00	$2.04	$43.64
JLCPCB	$497.06	$50.30	Historical - v1 receipt	0.35	$173.97	$38.52	$759.85
DigiKey	$58.14	$8.49	Historical - v1 receipt	0.08	$4.65	$4.51	$75.79
LCSC	$80.35	$22.21	Estimated	0.35	$28.12	$6.23	$136.91
Jameco	$67.85	$8.00	Estimated	0	$0.00	$5.26	$81.11
BatteryMart	$43.90	$6.00	Estimated	0	$0.00	$3.40	$53.30
SparkFun	$31.50	$8.00	Estimated	0	$0.00	$2.44	$41.94
Prusa Research	$54.99	$10.00	Estimated	0	$0.00	$4.26	$69.25
Walmart	$9.99	$0.00	Estimated	0	$0.00	$0.77	$10.76
Ted Pella	$30.00	$12.00	Estimated	0	$0.00	$2.33	$44.33
Isolate It	$35.00	$8.00	Estimated	0	$0.00	$2.71	$45.71
Carrier import processing fee (PiezoDrive)	$0.00	$0.00	Estimated		$0.00	$0.00	$65.00
ALL ORDERS	$1,112.07	$214.26			$219.84	$86.19	$1,697.36

In-kind and campus (not in the total)

Item	Treatment	Reference value if bought	Note
Pt/Ir tip wire	In-kind from a UCSD lab	$277.20	Reference: Bruker DPT10 10 cut tips $277.20; Advent 0.25 mm x 0.2 m wire $297.00. Jacob: sourced from UCSD.
HOPG sample	In-kind from a UCSD lab	$77.00	Reference: NanoAndMore ZYH 10x10x1 mm $77.00. Jacob: sourced from UCSD.
Machining	Not needed		All structure is 3D printed; no machined parts, no shop fees.
3D printer time	Campus (assumed)		EnVision has 3D printers; fees not stated and Jacob's eligibility unconfirmed. Filament is purchased (ME-01).
Granite base or optical breadboard	Campus (assumed)	$75.15	Reference if bought: Grizzly G9649 $75.15 + freight.
Bench power supply (+/-15 V class)	Campus (assumed)		v1 bought a $42.74 supply.
Oscilloscope, multimeter, soldering station, hot air	Campus (assumed)		EnVision lists soldering stations and oscilloscopes.
Silver epoxy or silver paint for mounting HOPG	Campus (assumed)	$133.13	Reference if bought: MG 8331D-14G $133.13 (DigiKey via aggregator).
Wire (40 AWG magnet wire, 30 AWG Kynar), coax, hook-up wire	Campus (assumed)		Common lab stock.
Connectors, headers, DIP sockets, ribbon/USB/DB9 cables	Campus (assumed)		Common lab stock; v1 spent about $50 on these on Amazon.
Fasteners, springs, small magnets, BeCu or steel shim for clips	Campus (assumed)		Common lab stock.
Consumables: IPA, acetone, flux, solder, wipes, gloves, tape, copper tape	Campus (assumed)		Chemistry stockroom sells against a project number; v1 lessons (99% IPA, RMA flux, copper tape) still apply.

Item	Treatment	Reference value if bought	Note
Tools: flush cutters, tweezers, ESD strap	Campus (assumed)		
Shield enclosures	Campus (assumed)		Die-cast box or copper tape from lab stock.

Requirements checklist

ID	Requirement	Source	Where it is	Status
R01	Total under $2,000	Jacob	Budget summary	See total vs ceiling
R02	Built at UCSD; campus items listed, not hidden	Jacob	In-kind and campus tab	Covered (assumed, unverified)
R03	No reuse of v1 parts	Jacob	All lines are new purchases; JLCPCB order is a new order	Covered
R04	Tip wire and HOPG from UCSD; printed structure; v1-style JLCPCB boards; filament bought	Jacob	In-kind tab; ME-01; EL-01..04	Covered
R05	Better piezo, possibly a tube (Vasquez), not a confirmed diagnosis	Vasquez	PZ-01, Piezo options	Covered
R06	Piezo dependencies priced (driver, power, extra channels)	Jacob	HV-01..04, Piezo options	Covered; driver provisional
R07	Sample plate replaced, not v1's springs; coarse approach stays a stepper (Jacob)	Jacob	PZ-02..04, ME-04, ME-05, Approach options	Covered
R08	Every purchased line has a direct link or says why not	Jacob	Detailed BOM 'Source link'	Covered; 4 lines explain missing direct links
R09	Temperature monitoring; enclosure	Zahl	PA-05; enclosure from campus stock	Covered
R10	No plastics in the mechanical loop	Zahl	Conflicts with the printed structure	OPEN - flagged (U4)
R11	Steel balls at pivots; stiff screw drive	Zahl	ME-04; PZ-04	Covered
R12	1 GOhm feedback + ~1.6 pF; batteries; ceramic caps; shielding	Baird	PA-01, PA-03, PA-04; caps on rev 2 board; shield from campus	Covered
R13	Cut Pt/Ir tips; HOPG on conductive epoxy / steel disc	Baird	In-kind wire/HOPG/epoxy; ME-05	Covered
R14	Scope the preamp output	Baird	Campus oscilloscope (assumed)	Campus, unverified
R15	Shipping, tariffs, tax, contingency, pack sizes	Jacob	Vendor-order totals; Budget summary	Covered
R16	Earlier full-build scenario kept for comparison	Coordinator	Budget summary + STM_v2_budget_full_build.xlsx	Covered

v1 invoiced orders, with shipping and tax allocated by line value

Order	Date	Item	Qty	Unit	Line merchandise	Alloc. discount	Alloc. shipping	Tariff / import tax	Alloc. sales tax	Landed line cost
JLCPCB	2026-07-15	2-layer bare STM controller PCB	3	$1.68	$5.04	-$0.21	$0.52	$1.69	$0.30	$7.35
JLCPCB	2026-07-15	Bare preamp PCB (OPA627)	3	$0.80	$2.40	-$0.10	$0.25	$0.81	$0.14	$3.50
JLCPCB	2026-07-15	Assembled controller board SMT assembly, v1 controller design	2	$209.64	$419.27	-$17.22	$43.30	$140.73	$25.14	$611.22
JLCPCB	2026-07-15	Assembled preamp board SMT assembly, v1 preamp rev 2	2	$30.17	$60.34	-$2.48	$6.23	$20.25	$3.62	$87.96
Thorlabs	2026-06-24	FAS200 2 in ultra-fine adjustment screw 1/4-80	3	$8.78	$26.34	$0.00	$15.26	$0.00	$1.65	$43.25
DigiKey	2026-07-18	LT1469IN8#PBF dual op-amp	2	$16.35	$32.70	$0.00	$7.77	$2.62	$2.24	$45.33
DigiKey	2026-07-18	DB9 receptacle A-DF 09 A/KG-T4S	4	$0.76	$3.04	$0.00	$0.72	$0.85	$0.21	$4.82
DigiKey	2026-08-02	Keystone 11311	2	$2.81	$5.62	$0.00	$8.49	$0.45	$0.38	$14.94
DigiKey	2026-09-09	Keystone 11301	2	$2.09	$4.18	$0.00	$8.49	$0.75	$0.31	$13.73
Invoiced orders total					$558.93					$832.10
Check vs invoice totals ($832.10)										$0.00

The Amazon purchases (42 small orders, shown as a total only) and the full source list are in the spreadsheet.